they will collide. Notice, however, that if these same point particles are randomly rolling around on a two-dimensional plane such as the spatial extent of Flatland, it is likely that they will never collide. The second spatial dimension opens up a new world of trajectories for each particle, most of which do not cross each other at the same point at the same time. In three, four, or any higher number of dimensions, it gets increasingly unlikely that the two particles will ever meet. Brandenberger and Vafa realized that an analogous idea holds if we replace point particles with loops of string, wrapped around spatial dimensions. Although it's significantly harder to see, if there are three (or fewer) circular spatial dimensions, two wrapped strings will likely collide with one another—the analog of what happens for two particles moving in one dimension. But in four or more space dimensions, wrapped strings are less and less likely ever to collide—the analog of what happens for point particles in two or more dimensions.4

And so we have the following picture. In the first moment of the universe, the tumult from the high, but finite, temperature drives all of the circular dimensions to try to expand. As they do, the wrapped strings constrict the expansion, driving the dimensions back to their original Planck-size radii. But, sooner or later a random thermal fluctuation will drive three dimensions momentarily to grow larger than the others, and our discussion then shows that strings which wrap these dimensions are highly likely to collide. About half of the collisions will involve string/antistring pairs, leading to annihilations that continually lessen the constriction, allowing these three dimensions to continue to expand. The more they expand, the less likely it is for other strings to get entangled around them since it takes more energy for a string to wrap around a larger dimension. And so, the expansion feeds on itself, becoming ever less constricted as the dimensions get ever larger. We can now imagine that these three spatial dimensions continue to evolve in the manner described in the previous sections, and expand to a size as large as or larger than the currently observable universe.

Cosmology and Calabi-Yau Shapes

For simplicity, Brandenberger and Vafa imagined that all of the spatial dimensions are circular. In fact, as noted in Chapter 8, so long as the circular dimensions are large enough that they curve back on